

World Health Organization Webinar – Monday, November 24th

From Data to Impact: Advancing Healthcare-Associated Infection Surveillance for Safer Care and a Healthier Future

Event topics

- I. WHO global strategy
- II. HAI and AMR surveillance methods and elements
- III. HAI and AMR impacts across the globe
- IV. Country examples

Speakers – WHO directors from various world regions

[Amal Al-Maani](#) – Director of Central Dept. for Infection Prevention and Control in Oman

Paul Rogers – Infection Prevention Program Manager

[Yvan Hutin](#) – Director of Universal Health Coverage Dept. in Eastern Mediterranean Region

Benedetta Allegranzi – Infectious Diseases Specialist, Switzerland

Maha Talat – Regional Advisor, Ontario

Sara Tolba – Medical Epidemiologist

Mandy Deeves – Control Technical Officer

Focus items:

- I. Key themes and takeaways about HAI surveillance systems.
- II. How AMR surveillance integrates into infection prevention strategies.
- III. Global and national-level implementation challenges or examples discussed.
- IV. Any data visualization, analysis, or health data application methods mentioned.

HAI - World AMR week

- Disproportionally affect low-income countries
- Most cases preventable
- Place strain on health systems
- Data is key for addressing gaps
- Admission without the infection or incubation of it
- About 45 different types of HAI

WHO core component 4: HAI and AMR surveillance

WHO IPC Global Action Plan and Monitoring Framework

- Minimum requirements build on full requirements
- NEW global strategy: Section 5 – data for action
 - o Implement efficient data collection and analysis training
 - o Follow up with frequent and consistent collection
- “By 2030 everyone accessing healthcare is safe from HAI”
- WHO advises to establish national HAI and AMR surveillance systems
 - o Why?
 - Early warning to detect epidemic and pandemic-prone pathogens
 - For monitoring antimicrobial consumption
- Goals for national surveillance but also facility-level
 - o As of 2023-2024 50% of facilities had some form of surveillance

Trends

WHO HAI Surveillance Practical Documents

- Most HAIs caused by antimicrobial resistant organisms
 - o Always been a close link
- Surveillance
 - o Aid in early detection, with signs symptoms to assign
 - o Identify causative organisms and their antimicrobial susceptibility profiles
- Methods of HAI surveillance
 - o Active vs passive
 - o Pt based vs lab based
 - o Comprehensive vs
 - o Incidence vs prevalence
 - Means rate vs proportion
 - Signify one as dedicated trackable metric
- Identifying types of HAIs
 - o Core: BSI, UTI, SSI, PNA
- Globally: WHO advises for harmonized case definitions to ensure consistent identification of HAI diagnostic criteria
- What’s involved?
 - o Data collection
 - Standardized
 - Sources: pt charts, electronic health records,
 - Automated data entry IT systems preferred to reduce errors
 - Improved accuracy when using algorithms
 - o Data analysis
 - Incidence, prevalence, device associated rates used to summarize HAI data
 - o Data interpretation
 - Turning raw data into meaningful recommendations
 - Where does infection happen and what to do now?

- Identify high risk groups & implications for risk planning
- Data communication
 - Continuous accountability and improvement
 - Share surveillance reports & aim for stakeholder engagement

Integration opportunities

- AMR surveillance:
 - Diagnostic cultures
 - Labs generate antibiograms
 - Data entered into standardized systems
 - Typically passive, based on clinician requested samples
- “Surveillance is not a recommendation, but a necessity”

AMR vs HAI Surveillance – Complementary but Fragmented

Key differences

	HAI Surveillance	AMR Surveillance
Aim 🎯	Inform IPC policies & interventions	Guide empiric treatment & monitor resistance trends
Focus 👥	Patients with HAIs (healthcare-associated pathogens)	Both HAI and CAI pathogens (but differentiation must improve)
Data 📊	Clinical, epidemiological, procedures (often missing microbiological data)	Laboratory microbiology + AST (often missing demographics/clinical info)
Platform 🌐	WHO IPC/HAI frameworks	WHO GLASS
End-users 👤💻	IPC teams, facilities, regulators	Policymakers, prescribers, One Health sectors
Limitations ⌚	Limited lab capacity; may emphasize MDROs → bias	Mostly retrospective; limited IPC value

- High-income countries (EU, USA) track HAI and AMR surveillance simultaneously
- Improve collection of more granular individual level data
 - Infection origin
 - Other variables (PT type, hospital unit type)
- Next steps
 - Initial stages – allow integrated reporting
 - Deep dive technical discussions needed on
 - Collaborations
 - Current work and modification, including individual level data

Q&A panel

As more health systems move toward automated data-entry IT systems and algorithm-based detection, what safeguards are needed to ensure these advanced systems do not miss subtle clinical indicators of HAI or AMR infection (especially those that are not well documented in electronic patient charts)?